INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid-Spring Semester Examination 2022-23

Date of Examination: 17th Feb 2023 **Session: (FN/AN) AN** **Duration: 2 hrs.** **Full Marks: 50**

Subject No.: AI61004 **Subject: Statistical Foundations of Artificial Intelligence and Machine Learning**

Department/Center/School: ***Centre of Excellence in Artificial Intelligence***

Specific charts, graph paper, log book etc., required: None

Special Instructions (if any): PLEASE SHOW ALL STEPS IN DETAILS (NOT JUST FINAL ANSWER)

ANSWER ANY 5 QUESTIONS. PARTS OF SAME QUESTION MUST BE ANSWERED TOGETHER

Q1. i) State the axiomatic definition of probability, and define the distribution of a random variable with respect to this definition. Illustrate this using the case of a random experiment where two coins are thrown simultaneously. **[4 marks]**

ii) You are tracking the number of Whatsapp messages you receive, along with their arrival times (t_1, t_2,....). You observe that the number N_t of messages you receive in any time-interval of length t follows a Poisson distribution with parameter λ*t. Now, consider the inter-arrival times $d_1=t_2-t_1$, $d_2=t_3-t_2$, ... What is the PDF of these inter-arrival times? (Hint: get the CDF first) What is the expected inter-arrival time? **[4+2=6 marks]**

~~Q2. i)~~ For any random variable X with X~N(m, s^2), P(m<X<m+s)=0.34 and P(m+s<X<m+2s)=0.135 (roughly). Suppose you have 100 observations from a Gaussian distribution whose variance parameter is known to be 25, and the sample mean of the observations comes to 10. Specify a range, such that you are at least 95% confident that the true mean of the Gaussian must lie within that range. **[4 marks]**

~~ii)~~ You have a coin with bias 0.5, which you don't know and are trying to estimate by Maximum Likelihood. If you do 10 tosses, how likely is it, that your estimate from these tosses has an error at most 0.1? **[3 marks]**

ii) I have n observations from N(m, s^2). I am under the wrong impression that m=0, and I try to estimate s^2 accordingly. What will be the bias of my estimate? **[3 marks]**

Q3. i) Show that Gamma distribution is a Conjugate Prior for Poisson distribution. **[6 marks]**

~~ii) Find~~ maximum likelihood estimate of parameters of Uniform distribution U(a,b). **[4 marks]**

Q4. You want to fit a Gamma distribution on the daily rainfall (in mm.) observed in Kharagpur during Monsoon season. Based on the observations below, estimate the parameters of the Gamma distribution using Method of Moments. **[10 marks]**

D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
23	14	7	3	9	39	10	1	1	4
D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
5	5	14	7	4	3	9	3	15	3

Q5. Consider a Naïve Bayes Classifier over C classes using D dimensional features, where each feature can take any real value. Explain, with full calculations, how you can find the maximum-likelihood estimate of the parameters for each feature, if you represent them with Gaussian distributions. Derive a posterior on the NBC's prior distribution's parameters. **[7+3=10 marks]**

Q6. i) Explain how linear and ridge regression problems can be cast in a probabilistic setting, and how the posterior distribution on the coefficient vector can be estimated. **[4 marks]**

ii) Consider an autoregressive process where $X_t \sim N(aX_{t-1}, s^2)$. From N observations $X_1, X_2, \ldots X_N$, how can I make maximum-likelihood estimate of a and s^2? How can I make a Bayesian estimate of a, assuming s^2 is known? **[2+4=6 marks]**